

## **Data Structures Lab, B.Tech 2nd Semester**

### **Instructions**

1. This is only for practice. No need to submit it.
2. Complete it by 5:00 PM today. Your completion will be reviewed by the Teaching Assistants.

### **Practice Assignment 6**

1. (a) Write a menu driven program that gives three choices to user.

Choice 1: Enqueue

Choice 2: Dequeue

Choice 3 Exit

If the user selects choice 1, the user is asked for a number and it is added to the queue. After that again the same choices come.

If the user selects choice 2, one element is removed from the queue and is displayed. After that again the same choices come.

If the choice is 3, the program exits.

Implement queue using circular array.

- (b) Modify the above program such that a linked list is used instead of array for queue implementation.
- (c) Create a circular doubly linked list and insert a few values. Print content of the linked list from first node to last node. Also print contents in reverse order.
- (d) Write a program to implement a queue using two stacks.